Yuxin Jiang

Los Angeles | +1 213-245-6579 | annajiang2178@gmail.com | Google Scholar link

EDUCATION

University of California, Los Angeles, USA | *Ph.D. in Computer Science* 2026.09-present

Imperial College London, UK | *B.Sc. in Mathematics with Statistics for Finance* 2020.10-2023.08

- Wine Rating Project**:** Employed PCA and Random Forest to classify wines, optimizing tree depth and achieving 98% acc.
- 3D Metaverse VR Project: Implemented CNN for face detection, effectively separating individuals from the background.

SKILLS

4 years of industrial ML experience. 14 months experience in R&D of cutting-edge technologies in pre-training and post-training VLA, VLM, U-net and diffusion-based video generation models. Experience in training 3B+ models in hundreds of GPUs, applying models to improve product performance, preprocessing data for training and optimizing inferencing and training speed.

- Domains: VLM, Video Generation, NLP, CV, Deep Learning, Math Modeling, Quantitative Trading, Web/Mobile development, UI/UX
- Programming Languages: Python, PyTorch, TensorFlow, Scikit-learn, R, SQL, Javascript, HTML/CSS, Vue, Node.js, Flask, Docker, UE5.
- Specialization: Deep Learning Specialization, Machine Learning Specialization from DeepLearning.AI

WORKING EXPERIENCE

Nirvana Robotics, USA (Remote) | *Research Collaborator* 2026.01-present

Remote research collaboration on world models and agentic policy generation for robotic manipulation

- Lead on **world-model memory** research, co-developing **MemoryVAM** (first author), an episodic memory mechanism that conditions video-world-model policies on episode history and enables **long-horizon manipulation** beyond a short observation window.
- Lead to an **agentic policy generation** project for robotic manipulation, in which language-driven agents synthesize, verify, and refine executable control policies for real-robot deployment.

UCLA Artificial Intelligence & Visual Computing Laboratory @ University of California, Los Angeles, USA. 2025.09-2026.07

Visiting Researcher | Advisor: Prof. Chenfanfu Jiang

- Work and research on **generative AI, world models**. Proposed Research topic: Physically Reliable World Model.

Shanghai Zhiyuan Xinchuang Technology Co., Ltd. (formerly Shanghai Agibot Innovation Technology Co., Ltd.; renamed 07/2025), Shanghai | *AI Research Institute* | ML Researcher Intern with return 2024.06-2025.08

- Core member in Project EVAC, conducted research and built an **action-conditioned world model (EVAC)** based on a pretrained **video generation model** for robotics action policies to serve as a data engine and an evaluator with 2 pubs.
- Core member in Project AGIBOT WORLD, integrated a latent planner with **large foundation models** for embodied agents with 2 pubs.
- Core member in **post-training** pretrained models for multi commercial-use tasks by integrating multi-modal input, processed data etc.
- Core member in **data-processing** for pre-training and post-training.

Ant Cloud Technology Group, Shanghai | *AI Digital Human Institute* | ML Engineer & Technical PM Intern 2024.01-2024.05

- Integrated **Automatic Speech Recognition**, **chatbots** combined with a local knowledge graph, **Text-to-Speech** models like **GPT-SOVITS** and **CHATTTS**, and **real-time lip-sync** within **UE5**, using Blueprints to build a 3D AI digital human for commercial use and demos.

ACADEMIC PUBLICATIONS

MemoryVAM: Integrating Memory into Video Action Model for Robot Manipulation (Conference on Robot Learning (CoRL) 2026, accepted; RSS 2026 Workshop on Robot World Models, accepted, First Author)

- Built an **episodic memory mechanism** for video-world-model policies via a **Recap-Cue module**, where a Perceiver-based compressor maps per-frame CLIP embeddings into compact **memory tokens** and a lightweight Cue Gate estimates task completion from memory and language.
- Injected memory tokens into both the **video backbone and action decoder** (portable across U-Net and DiT by changing only the cross-attention interface), improving LIBERO-Mem average success from **5% to 42.5%** and reaching **78.3% / 80.0% / 75.0%** success on real-robot counting, spatial recall, and sequential tracking.

[Genie Envisioner: A Unified World Foundation Model for Robotic Manipulation]() (ICLR 2026)

- Used a large-scale, **instruction-conditioned DiT video diffusion** model that captures the spatial, temporal, and semantic dynamics of real-world robotic interactions in a latent space and a **flow-matching action decoder** to generate actions.

[RoboEdit: Turning Human Manipulation Videos into Scalable Robot Experience]() (arXiv preprint, 2026)

- Developed a **human-to-robot video editing suite** that converts human manipulation videos into action-consistent, physically plausible robot videos with aligned **3D hand states**, unlocking abundant unlabeled human video for robot training.
- Built **RoboEdit-ADC**, an automatic pipeline reconstructing and retargeting 3D interactions from RGB video across embodiments, generating **RoboEdit-14M** (174K aligned video pairs, 14M frames, seven robot embodiments) to supervise downstream control policies.

[FetchMan: Learning Visual Humanoid Loco-Manipulation Policies from Simulated Experiences]() (RSS 2026 Workshop on Whole-Body Control and Bimanual Manipulation, accepted)

- Built an end-to-end **sim-to-real pipeline** spanning over **150,000 scenes**, showing that behavior cloning on synthetic demonstrations plateaus and that refining the cloned policy with **Flow-GRPO** on a single sparse reward breaks through that ceiling.
- Deployed the policy **zero-shot on a real Unitree G1** humanoid, walking to and grasping targets across unseen scenes at **73.3% success**, and released the FetchMan-Bench simulation benchmark.

[SUN: Persistent Programs For Language-Grounded Control-to-Learning-to-Real Policies]() (arXiv preprint, 2026)

- Introduced **Semantically UNified (SUN) Programs**, typed executables in which geometric and contact relations are defined once and compiled into aligned **MPC costs, RL rewards**, satisfaction predicates, transition guards, and diagnostics.
- Synthesized SUN Programs automatically from language and scene semantics using **large vision-language models**, achieving **82.03% macro-success** across nine tasks versus 35.67% for sparse-reward and 24.75% for Stage-BC baselines.

[Enerverse-ac: Envisioning embodied environments with action condition]() *(NeurIPS 2025 Embodied World Models Workshop, Oral & Outstanding Paper Award, First Author)*

- Designed EVAC achieving **video generation** via trajectory control by **cross-attention and channel-wise concatenation** through U-net.
- Developed a pipeline to serve EVAC as a data engine and evaluator, augmented human-collected trajectories into diverse datasets and to **post-train** Go-1 model increase acc. by 8% and generates realistic video observations for **consistent policy testing**.

[AgiBot World Colosseo: A Large-scale Manipulation Platform for Scalable and Intelligent Embodied Systems]() *(IROS 2025, Best Paper Award Finalist; IEEE Transactions on Robotics, 2026, Author for pretraining)*

- Extracted **K-V cache from VLM** to enrich semantic understanding. Additionally, developed a unified **transformer-diffusion decoder** facilitates latent action generation by **pre-training** a latent planner and final action decoder.
- Utilized large-scale open-source datasets to **pre-train** a **VQVAE latent model** to enhance the model's generalizable ability.

[Genie Centurion: Accelerating Scalable Real-World Robot Training with Human Rewind-and-Refine Guidance]() (arXiv preprint, 2025)

- Implemented a pipeline by human intervening or robo-guided functions during data collection in response to model inference failures to improve data usability and reduce data required in **finetuning**.